TOPIC 2 · PERSONALISATION, .RULES AND SKILLS FOR EXCEL
LAB 6 THEME · REPORTING SKILL

Lab 6 — Build a Maintenance Report with Two Skills

Maps to	LO2: distinguish account Personalisation, workbook .Rules and reusable Skills; configure, apply and verify each using water-service evidence.
Duration	30 minutes
Tools	Excel and PowerPoint Copilot · Microsoft 365 Copilot web · OneDrive (Skills folder only)
Mock data	data/Northstar_Work_Orders.xlsx
Lab files	data/templates/maintenance-reporting/SKILL.md (Reviewed reporting Skill fallback)
Course	GenAI for Engineers · v2.4
Prompts	https://tertiarycourses.github.io/IES-GenAI-for-Engineers/prompts/#lab-6

Goal

Install two skills, water-service-pareto and maintenance-reporting, and use them together to build a maintenance report in Excel, then use the reporting skill in PowerPoint.

What You Will Build

Maintenance_Summary with a failure-mode Pareto, Assignment_Proposals, and an improved Northstar_Initial_Update.pptx.

Prerequisites

- Labs 1–2 verified brief and four-slide update; Lab 5 Skill-authoring method.
- Desktop Excel on Windows or Mac. Download this lab folder and open the files straight from its data folder; save any new file in the same folder. All data is fictional.

Data note

Northstar Water Services is a fictional water-pipe maintenance and service company. All service hubs, staff and work orders are synthetic. Use the fixed reporting date 2026-09-30; missing facts remain UNKNOWN. Proposals do not change real assignments. Training email drafts are never sent.

Steps

Step 1. Upload the skill to OneDrive. In Excel open Copilot and select Settings (...) > Manage plugins & skills > Custom skills. Turn Custom skills on and select Add in OneDrive (or Create OneDrive folder, then Open skills folder). OneDrive opens at your skills folder. Select Create or upload > Folder upload and choose the water-service-pareto and maintenance-reporting folders in this lab's data/templates folder.

PROMPT / CHECK — copy and adapt

```
OneDrive : Documents > Copilot > Microsoft Excel > skills > water-service-pareto > SKILL.md
OneDrive : Documents > Copilot > Microsoft Excel > skills > maintenance-reporting > SKILL.md
```

Step 2. Turn the skills on. Back in Excel select Settings (...) > Manage plugins & skills > Custom skills > Refresh, then turn water-service-pareto and maintenance-reporting on. Close the dialog and type @ in the Copilot prompt box: water-service-pareto and maintenance-reporting should appear. If not, wait a minute and select Refresh again.

PROMPT / CHECK — copy and adapt

```
Check : water-service-pareto appears when you type @
Check : maintenance-reporting appears when you type @
```

Step 3. Build the summary. Open data/Northstar_Work_Orders.xlsx, select New chat, choose Allow editing and send the prompt. Select Done to keep the change, or Undo to reverse it. You should see 120 total, 48 closed, 12 cancelled, 60 backlog, 50 overdue, on-time 30/48 = 62.5% and cost S$35,827.50.

PROMPT / CHECK — copy and adapt

```
EXCEL - ALLOW EDITING:
@maintenance-reporting Build a Maintenance_Summary sheet for 30 September 2026 with the total, closed, cancelled, backlog, overdue, on-time closure rate and cost.
```

Step 4. Add the failure-mode Pareto. Send the prompt and select Done. The chart is sorted from highest to lowest: Corrosion 42, Joint seal leak 30, Valve seat wear 22, Pump bearing wear 10, Pressure surge 8, Meter fault 4, Blockage 2, Third-party damage 2.

PROMPT / CHECK — copy and adapt

```
EXCEL - ALLOW EDITING:
@water-service-pareto Make a Pareto chart of Failure_Mode for all work orders and place it on the Maintenance_Summary sheet, to the right of the numbers.
```

Step 5. Suggest owners. Send the prompt and select Done. Assignment_Proposals lists 6 jobs, each marked as a proposal, and Assigned_To on Work_Orders is unchanged. Save the workbook.

PROMPT / CHECK — copy and adapt

```
EXCEL - ALLOW EDITING:
@maintenance-reporting Add an Assignment_Proposals sheet for the backlog jobs with no owner. Suggest an owner from tblTechnicians with the same job category and free hours, and mark each one as a proposal. Don't change Assigned_To.
```

Step 6. Use the skill in PowerPoint. Open data/Northstar_Initial_Update.pptx. In Copilot select + > Choose skills > Manage skills > Add skill and upload data/templates/maintenance-reporting/SKILL.md, then send the prompt. Every number must stay the same.

PROMPT / CHECK — copy and adapt

```
POWERPOINT COPILOT:
@maintenance-reporting Improve these 4 slides with clearer headlines and less text. Keep every number and keep 4 slides.
```

Copy buttons for every prompt: https://tertiarycourses.github.io/IES-GenAI-for-Engineers/prompts/#lab-6

Test It

Check	Expected result
Summary	120 total; 60 backlog; 50 overdue; 30/48 = 62.5%; S$35,827.50.
Pareto	Corrosion 42, Joint seal leak 30, Valve seat wear 22, Pump bearing wear 10, Pressure surge 8, Meter fault 4, Blockage 2, Third-party damage 2.
Proposals	6 proposals; Assigned_To unchanged.
PowerPoint	4 slides, same numbers, clearer headlines.

Troubleshooting

If this happens	Fix
Copilot cannot see my file	In Excel, save the file and open Copilot again. In Word, PowerPoint or Outlook, upload the file with + > Upload images and files, and upload it again after you change it.
A number is different from the lab	Clear any slicers or filters, check the table has no extra rows, then ask again in a new chat.
A button or feature is missing	Check you are signed in with your course account and Microsoft 365 is up to date, then tell your trainer.

Challenge

Open a new blank workbook and ask @maintenance-reporting for last month’s report: a good skill asks for the data instead of inventing numbers.

Reflection

Which evidence proves the output is correct, and which decision still needs a person?